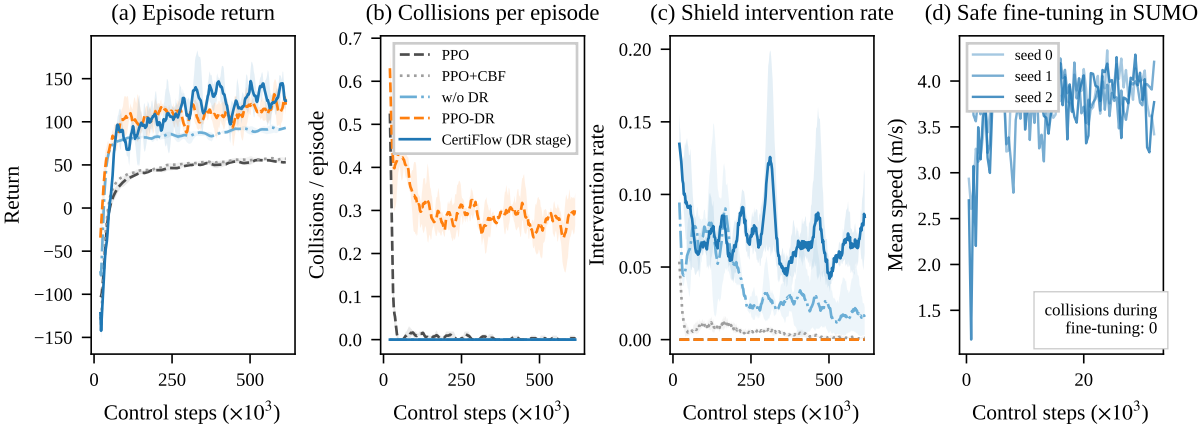

(a) Episode return
Return
150
100
50
0
−50
−100
−150
0
250
500
Control steps (×10^3)
(b) Collisions per episode
Collisions / episode
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
PPO
PPO+CBF
w/o DR
PPO-DR
CertiFlow (DR stage)
0
250
500
Control steps (×10^3)
(c) Shield intervention rate
Intervention rate
0.20
0.15
0.10
0.05
0.00
0
250
500
Control steps (×10^3)
(d) Safe fine-tuning in SUMO
Mean speed (m/s)
4.0
3.5
3.0
2.5
2.0
1.5
seed 0
seed 1
seed 2
collisions during
fine-tuning: 0
0
20
Control steps (×10^3)